

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: The Internet & WWW

Introduction

- The Internet is the global network of interconnected computers using the TCP/IP protocol suite; the World Wide Web (WWW) is the system of interlinked documents (web pages) accessible over the Internet.

How the web works

- Data is moved in packets; each host has an IP address; the Domain Name System (DNS) translates names (www.example.ng) into IP addresses. Browsers (Chrome, Edge, Firefox) request pages from web servers over HTTP/HTTPS. HTML builds the pages; search engines (Google, Bing) index the web.

Services and components

- E-mail, file transfer, the web, social media, instant messaging, e-commerce, e-banking, video conferencing, cloud computing. A URL: scheme://domain/path. Browser tools: address bar, tabs, bookmarks, history, downloads, cache and cookies.

CLASS EVALUATION

1. Differentiate the Internet from the WWW.
2. What does DNS do?
3. Name three Internet services.
4. State the parts of a URL.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: E-Mail & Communication (Formal Practice)

Introduction

- E-mail remains the standard for formal electronic communication — applications, enquiries and official correspondence. This week practises professional e-mail writing.

A professional e-mail

- A clear subject line; a polite salutation (Dear Sir/Madam); a short first sentence stating the purpose; the body in brief paragraphs; a courteous closing (Yours faithfully / Regards); your full name, and signature block (name, school, phone). Attach files with proper names and sizes; proofread before sending; and use Reply vs Reply All carefully (Reply All can expose private addresses).

Etiquette and security

- Be brief and specific; avoid slang, emojis and capitals in formal mail; never forward chain letters or hoaxes; check attachments for viruses; never send personal data (NIN, PIN, passwords) by mail; log out of public computers; and remember that e-mail is not private — think before you write.

CLASS EVALUATION

1. Write a formal e-mail applying for a place in a summer ICT class.
2. What belongs in the signature block?
3. Why must you avoid 'Reply All' carelessly?
4. State two security rules for e-mail.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: Networks & Network Types

Introduction

- A computer network is a group of computers connected to share data, hardware (printers) and software, and to communicate. Networking is the backbone of modern computing.

Types by size

- **PAN** (Personal Area Network) — a person's devices (phone, laptop, watch) over Bluetooth; **LAN** (Local Area Network) — a room, school or office (Wi-Fi, Ethernet); **MAN** (Metropolitan Area Network) — a city (cable networks); **WAN** (Wide Area Network) — countries/continents — the Internet itself.

Topologies

- **Bus** — all nodes on one cable; cheap, but a break stops the network. **Star** — all nodes to a central switch/hub; easy to manage, but the hub is a single point of failure. **Ring** — nodes in a closed loop; data passes each node. **Mesh** — every node connected to every other; reliable but costly. **Hybrid** — combinations (most real networks).

Components and the peer-to-peer model

- Nodes (computers), servers and clients, NICs (network interface cards), switches, routers, hubs, cables and wireless access points; network operating systems. In peer-to-peer (P2P) each computer can act as both client and server (file sharing); in client-server, central servers provide services (the school's network).

CLASS EVALUATION

1. Define a computer network.
2. Differentiate LAN from WAN.
3. State one advantage and one disadvantage of the star topology.
4. What is the difference between a client and a server?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: Web Design Basics

Introduction

- Web design is the art of planning, building and maintaining web pages. A web page is written in HTML (structure), styled with CSS (presentation) and made interactive with JavaScript (behaviour).

HTML fundamentals

- Tags and elements: `<h1>`, `<p>`, `<a href="#">`; headings –; paragraphs ; links ; images ; lists `<ul>` /; tables ; forms , ; attributes give extra information (src, href, alt). Web pages must be saved with .html and opened in a browser.

CSS basics

- CSS controls colours, fonts, spacing and layout: selectors (element, class `.x`, id `#x`), properties and values; internal (in `<style>`), external (a .css file linked) and inline styles. Simple rules: `body { font-family: Arial; background: white; }`

Good design principles

- Clear structure and headings; contrasting colours and readable fonts; consistent navigation; appropriate images; fast loading; working links; mobile-friendly (responsive) layouts; accessibility (alt text, contrast); and original, correct content.

CLASS EVALUATION

1. Write the HTML of a page with a heading, a paragraph and a link.
2. What is the difference between HTML and CSS?
3. State three principles of good web design.
4. What does the tag's alt attribute do?

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WEEK: ELEVEN

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: Web Authoring Tools & Hosting

Introduction

- This week completes the web unit: the tools used to build pages (from a notepad to content management systems) and how a website is published on the Internet (domain, hosting and the file-transfer process).

Authoring tools

- Text editors (Notepad, VS Code) — full control by coding; WYSIWYG editors (Dreamweaver, WordPress) — design visually; CMS platforms (WordPress, Wix, Blogger) — templates, plugins and hosting built in; and website builders for e-commerce (Shopify). Choose by skill, purpose and budget.

Hosting and domains

- A domain name is the site's address (www.myschool.ng), leased from a registrar; web hosting is storage on a server — free (GitHub Pages, Netlify) or paid (shared, VPS, dedicated). Publishing: upload files to the server by FTP (FileZilla) or the hosting panel; the DNS record points the domain to the server.

Publishing process

- Design and code locally → test in a browser → validate → upload → test online → update regularly. Web standards (W3C) and site security (HTTPS with SSL certificates) are essential.

CLASS EVALUATION

1. Name three web authoring tools.
2. Differentiate between a domain name and web hosting.
3. What is used to upload files to a server?
4. Why is HTTPS important for a website?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS2

SUBJECT: Digital Technology

TOPIC: Computer Security & Cyber Safety

Introduction

- Computer security protects data and systems from theft, damage and misuse; cyber safety is the habit of using the Internet and devices safely. As the world goes digital, security is everyone's responsibility — especially the young.

Threats

- **Malware** — viruses, worms, Trojans, ransomware, spyware and keyloggers; **phishing** — fake emails, SMS and sites that steal passwords and card details (the 'Yahoo' scams and their modern forms); **social engineering** — manipulation by phone, WhatsApp, Facebook; **identity theft**; **hacking and DDoS attacks**; and **physical threats** — theft, fire, floods and power surges.

Protection measures

- Use strong, unique passwords (12+ characters with numbers and symbols) and two-factor authentication; install antivirus and antimalware software and keep it updated; update the OS and apps (patch the holes); avoid opening suspicious links, attachments and pop-ups; use HTTPS; back up data (the 3-2-1 rule: 3 copies, 2 media, 1 off-site); guard the SIM, phone and PIN; and lock the screen when away.

Cyber safety for students

- Protect personal information (name, address, school, bank); think before posting (the Internet never forgets); avoid befriending strangers online; report cyberbullying to parents, teachers and the police; never share passwords with friends; and use the reporting channels (the NITDA and the police's cybercrime desk).

CLASS EVALUATION

1. Define computer security.
2. Name five types of malware.
3. What is phishing?
4. State four measures that protect a computer.
5. Give three cyber safety rules for students.

